

Specifications for multi-organ-on-chip system for Microbiome and Metabolism Research Unit, USDA-ARS, Little Rock, AR

An organ-on-a-chip technology platform that will leverage multiple organ systems in the chip and will be used to perform biological assays to understand the biological processes and interactions between organs (such as gut-brain, gut-liver-brain, gut-liver). This system will be an initiative to reduce the use of animals in research by leveraging microphysiological systems (MPS) integrated into chips. This system will be used to understand how different microbiota, metabolites, and dietary components affect the gut, liver, and brain, and how they interact with one another. The complexity of this system is more relevant to the study of complex human physiology than simpler tools, such as cell culture.

Minimum Technical Requirements for the Multi-organ-on-a-chip control unit for optimal operation of the on-chip pumps

Quantity - 2

Dimensions: 251.2 mm x 206 mm x 150.2 mm each, benchtop compatibility

Weight of instrument: Not exceeding 4.8 kg

Acoustic profile: Sound power level during operation must be < 45 dB (A)

Pump and Flow control: must include user-controlled settings for fluid flow (speed and direction). Pump frequency must be adjustable from 0 to 2 Hz. Must include an on-chip micro-pump compatible with standard 24-well and 96-well inserts.

- a. **Capacity and Perfusion** - Must have the capacity to connect **4 to 8 chips** simultaneously. Must have the capacity to connect and perfuse at least **two cell types from two different tissues** in a single chip. Capacity to control up to 24 pneumatic actors
 - b. Must maintain growth of primary liver cells under conditions of 5% CO₂ and a flow rate of **0 µL/hour to 1000 µL**.
1. **Configuration:** Must be a “plug and play” configuration to accommodate different chips and modular expansion, allowing for installation by the end-user.
 2. **Imaging:** Must be compatible with **live tissue imaging** and designed for non-invasive online controls (optical methods).
 3. **Interface:** Must include an integrated **touchscreen display (7”)** for digital pressure and vacuum supply monitoring.

Compatible Chip details

- a. 2-organ-chip (24 well) – total 20 pieces
- b. Two microfluidic circuits/chip
- c. On-chip micro-pump
- d. Ready to use with 96-well and /or 24-well inserts
- e. Compatible with live tissue imaging

Other requirements:

1. Must include on-site installation
2. Must include customer support through phone calls and email

3. Product Compliance: Must be CE-certified and include a minimum 1-year full manufacturer's warranty.

Delivery to:

Microbiome and Metabolism Research Unit
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